CMSC 838B · STUDENT LECTURE · FALL 2026

Neural SDEs

Differentiating through stochastic dynamics

Running example: post-harvest ripening of 'Hass' avocado · *Xavier, Rodrigues & Silva, Foods 13:1150 (2024)*

Nishad Wajge · University of Maryland



Ripening stages 1–5. Fig. 1, Xavier, Rodrigues & Silva, Foods 13:1150 (2024), CC BY 3.0.

Xavier et al. (2024): Daily Images of 478 Avocados

Group	Storage	n	Stage 1→5 (fit)
T10	10 °C, 85% RH	192	17.6 d
T20	20 °C, 85% RH	143	8.5 d
Tamb	15.8–21.7 °C (mean 18.7), RH not logged	143	7.7 d

Images	Labels	Split
14,722 (2 sides × daily)	5-stage index; 10-stage refinement	70/15/15 by fruit

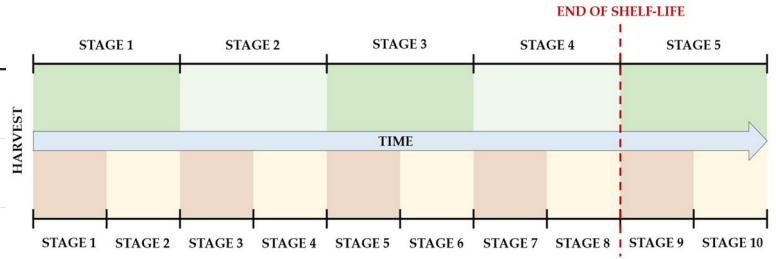


Fig. 2 (cropped): 5- vs. 10-stage index; end of shelf life = entry to stage 5. Xavier, Rodrigues & Silva, Foods 13:1150 (2024), CC BY 3.0.

Shelf life: **τ = first day at stage 5**

Stage-1→5 durations computed from the paper's Table 4 slopes (4 × |α|).

Three storage regimes, daily images, ordinal labels: an irregularly observed latent process.

RUNNING EXAMPLE

Their Pipeline: Classify, Then Map to Days Left



Stage accuracy (test, by picture)

Network	5-stage exact	±1	10-stage exact	±1
AlexNet	76.9%	99.4%	57.4%	92.6%
ResNet-18	78.4%	99.7%	60.8%	93.3%

Shelf-life MAE (days, 5-stage)

Group	true stage	predicted
T10	1.70	**1.82**
T20	0.73	0.75
Tamb	0.53	0.54
All	0.99	1.04

Tables 3 and 6, Xavier, Rodrigues & Silva, Foods 13:1150 (2024). 'Predicted' = by sample.

A strong per-image classifier; shelf life is a deterministic read-out of the current stage.

Their Shelf-Life Model, Read as a Constant-Drift ODE

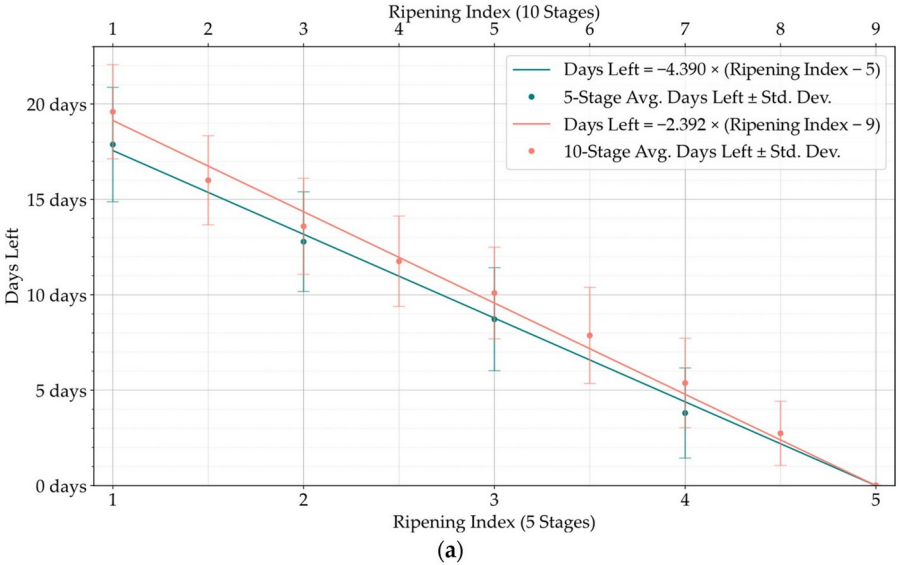


Fig. 6a (T10, cropped from Fig. 6): mean ± sd days left per stage. Xavier, Rodrigues & Silva, Foods 13:1150 (2024), CC BY 3.0.

$$\text{DaysLeft} = \alpha_g\,(s-5) \quad \Rightarrow \quad \frac{ds}{dt} \approx v_g = \frac{1}{|\alpha_g|}$$

Group	α (d / stage)	v = 1/|α|	R^2
T10	−4.390	0.228	0.953
T20	−2.116	0.473	0.963
Tamb	−1.929	0.518	0.959

α: Table 4 of the paper. v in stages / day (derived).

Missing: **dW** (spread around the line), memory of past images, uncertainty in τ.

The regression line is the drift; the error bars are the spread it leaves out.

A Point Estimate Hides Days of Spread

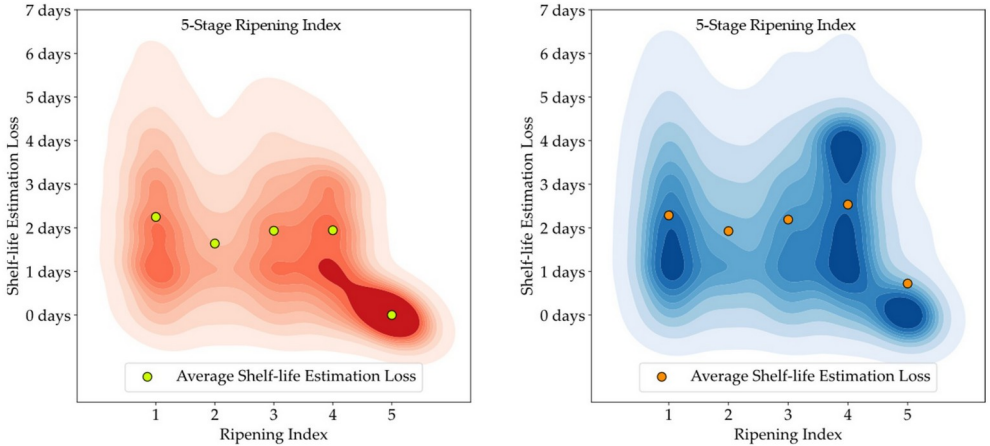


Fig. 11 (T10, cropped): shelf-life error vs. ripening index; left attributed labels, right predicted. Xavier, Rodrigues & Silva, Foods 13:1150 (2024), CC BY 3.0.

Group	|α| (d)	floor |α|/4	MAE	floor / MAE
T10	4.39	1.10	**1.70**	**65%**
T20	2.12	0.53	0.73	72%
Tamb	1.93	0.48	0.53	91%

MAE in days with attributed (human) labels, Table 6. Floor = MAE from position within a stage alone (uniform, width \|α\|).

Errors up to **~6 days** at stages 1–4 (T10); the paper reports only the point.

Stage quantization explains most of the error; the excess is largest at 10 °C, the longest horizon.

RUNNING EXAMPLE

What a Better Model Needs

Requirement	Per-image CNN (paper)	Sequence model	Latent ODE	Latent SDE
Use all past images	—	✔	✔	✔
Irregular / missing days	—	*partial*	✔	✔
Distribution over τ	—	*partial*	*partial*	✔
Temperature-structured rate	*partial*	—	✔	✔
Within-fruit noise (sd ∝ √h)	—	—	—	✔

Continuous time + intrinsic noise + structured drift → latent neural SDE.

Neural ODEs: Continuous-Depth Dynamics

residual block

$$h_{k+1} = h_k + f_\theta(h_k)$$

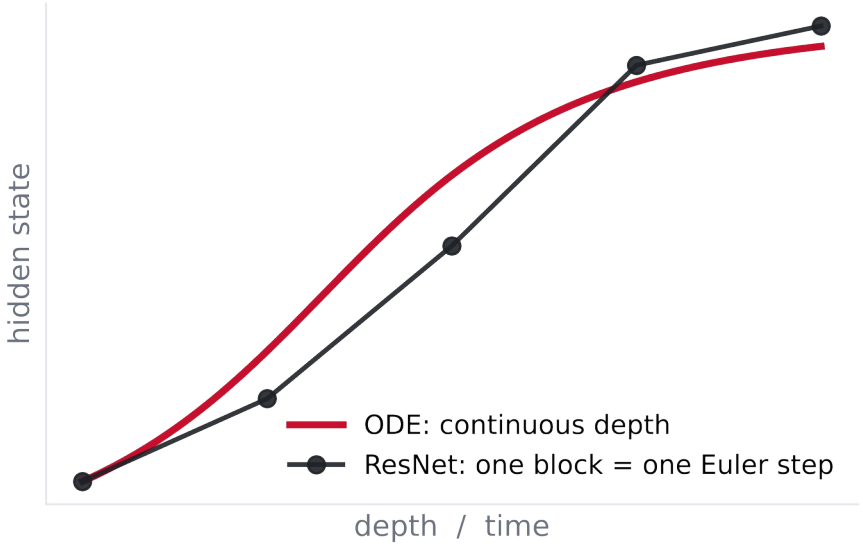
$\Delta t \to 0$

neural ODE

$$\frac{dh}{dt} = f_\theta(h, t)$$

latent version (paper = special case f = v_g)

$$\frac{dz}{dt} = f_\theta(z, t, c), \qquad z(0) = e_\phi(x_0)$$



ResNet = explicit Euler; a neural ODE delegates stepping to the solver.

Three Ways to Differentiate a Solver

Method	Memory (N steps)	Extra compute	Gradient	Diffrax
Backprop through steps	O(N)	≈ 1 backward pass	exact (discrete)	`RecursiveCheckpointAdjoint(checkpoints=N)`
Binomial checkpointing	O(log N)	O(N log N)	exact (discrete)	`RecursiveCheckpointAdjoint()`
Continuous adjoint	O(1) in N	backward solve, dim 2d + p	approximate	`BacksolveAdjoint()`

d = state dim, p = #params. Rows 1–2: discretize-then-differentiate. Row 3: differentiate-then-discretize.

Memory, compute and gradient exactness trade off; no method wins all three.

The Adjoint Method

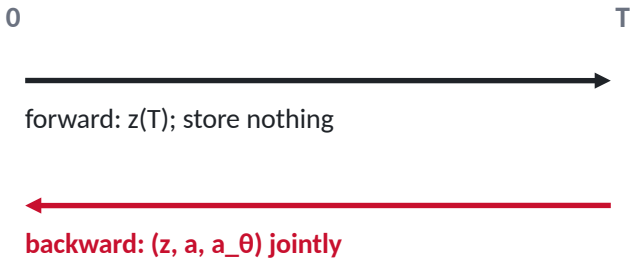
backward from T → 0 (derivation: A1)

state $\dot{z} = f_\theta(z, t)$

adjoint a = ∂L/∂z(t) $\dot{a} = -a^\top \partial_z f_\theta$

parameter gradient $\dot{a}_\theta = -a^\top \partial_\theta f_\theta, \quad a_\theta(T) = 0 \Rightarrow \frac{dL}{d\theta} = a_\theta(0)$

a(T) = ∂L/∂z(T)



Term	Primitive
$a^\top \partial f/\partial z$	VJP of f in z
$a^\top \partial f/\partial \theta$	VJP of f in θ

Reverse-mode AD in continuous time: one VJP per solver stage.

Itô Calculus: What You Need

SDE (integral form)

$$Z_t = Z_0 + \int_0^t \mu(Z_s, s)\, ds + \int_0^t \sigma(Z_s, s)\, dW_s$$

Brownian increments

$$W_t - W_s \sim \mathcal{N}(0,\ (t-s)I), \qquad (dW)^2 = dt$$

Itô's lemma

$$df(Z_t) = f'(Z_t)\, dZ_t + \tfrac{1}{2} f''(Z_t)\, \sigma^2\, dt$$

Itô → Stratonovich drift

$$\mu_{\text{Strat}} = \mu_{\text{Itô}} - \tfrac{1}{2} \sum_j (\partial_z \sigma_j)\, \sigma_j$$

Scheme	Calculus	Strong	Weak
Euler–Maruyama	Itô	½	1
Milstein	Itô / Strat.	1	1
Reversible Heun	Strat.	½	—

Full table: A7

Itô for modeling; Stratonovich for gradients (ordinary chain rule).

A Latent Neural SDE for Ripening



dynamics

$$dZ_t = \kappa(c)\,\mu_\theta(Z_t, t)\,dt + \sigma_\theta(Z_t, t, c)\,dW_t \qquad \kappa(c) = \exp\left[-\frac{E_a}{R}\left(\frac{1}{T_c} - \frac{1}{T_{\text{ref}}}\right)\right]$$

observation model

$$p(y \le k \mid Z_t) = \text{sigmoid}(c_k - q(Z_t)), \quad k = 1, \ldots, 4$$

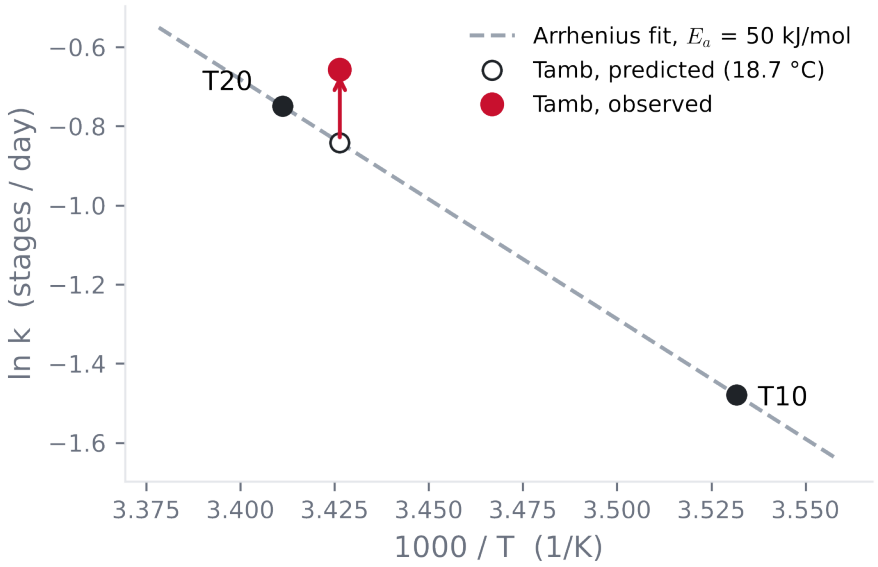
target

$$\tau = \inf\{t : q(Z_t) \ge b\}, \qquad \mathbb{P}(\tau \le t^* \mid x_{0:k}, c)$$

Paper	Here
ŝ from one image	filtered Z_t from all images
Days Left = α_g(ŝ − 5)	P(τ ≤ t* | images up to day k, c)

Drift = rate (physics prior × network); diffusion = fruit-to-fruit variability; τ = first passage.

Fitting the Drift Prior to the Paper's Rates



$$E_a = \frac{R \ln(v_{20}/v_{10})}{1/T_{10} - 1/T_{20}}$$

Quantity	Value
$Q_{10} = v_{20} / v_{10}$	2.07
E_a	50.4 kJ/mol
Tamb, predicted (18.7 °C)	2.32 d / stage
Tamb, observed	**1.93 d / stage**
Temperature implied	21.3 °C

Rates from the paper's Table 4; fit is ours (A5).

Two temperatures pin E_a; ambient ripens 20% faster than Arrhenius predicts → learned residual.

Do the Paper's Errors Have a Drift–Diffusion Shape?

first passage of drifted Brownian motion (A4)

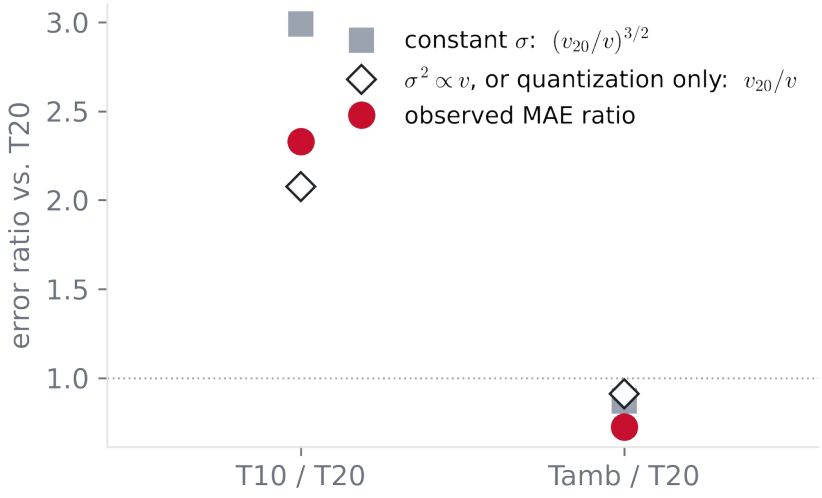
$$dX = v\,dt + \sigma\,dW, \qquad \tau = \inf\{t : X_t = L\}$$

$$\mathbb{E}\,\tau = \frac{L}{v}, \qquad \mathrm{sd}(\tau) = \sqrt{L\sigma^2/v^3}$$

$$\frac{\mathrm{sd}(\tau_{10})}{\mathrm{sd}(\tau_{20})} = \left(\frac{v_{20}}{v_{10}}\right)^{3/2}\ (\sigma \text{ fixed}), \qquad \frac{v_{20}}{v_{10}}\ \ (\sigma^2 \propto v)$$

Ratio	σ fixed	σ² ∝ v	quant. only	observed
T10 / T20	2.99	2.07	2.07	**2.33**
Tamb / T20	0.87	0.91	0.91	**0.73**

Observed = MAE ratios with attributed labels (paper, Table 6).



T10/T20 = 2.33 exceeds quantization alone (2.07): extra spread at 10 °C. Three numbers can't pin σ(T).

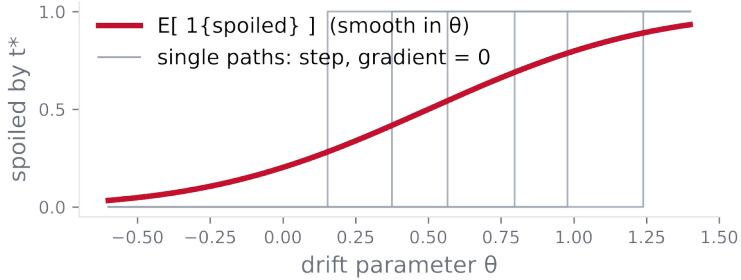
Pathwise vs. Score-Function Gradients

pathwise (reparameterization)

$$\nabla_\theta \mathbb{E}[f(Z_T)] = \mathbb{E}[\nabla_\theta f(Z_T(\theta, W))]$$

score function (likelihood ratio)

$$\nabla_\theta \mathbb{E}[f(Z)] = \mathbb{E}[f(Z)\,\nabla_\theta \log p_\theta(Z)]$$



Estimator for P(τ ≤ t*)	Bias	Variance	Notes
Pathwise on 1{τ ≤ t*}	**total (∇ = 0)**	—	fails
Sigmoid surrogate (ε)	O(ε)	↑ as ε → 0	needs ε schedule
Brownian-bridge crossing	low	low	smooth per step (A6)
Score function (Girsanov)	none	high	drift params only

Our target, P(τ ≤ t*), is discontinuous pathwise: the reparameterized gradient is zero.

The Stochastic Adjoint

backward from T, Stratonovich (∘), same W; μ = Stratonovich drift

$$da_t = -a_t^\top \partial_z \mu \, dt \; - \; \sum_j a_t^\top \partial_z \sigma_j \circ dW_t^j$$

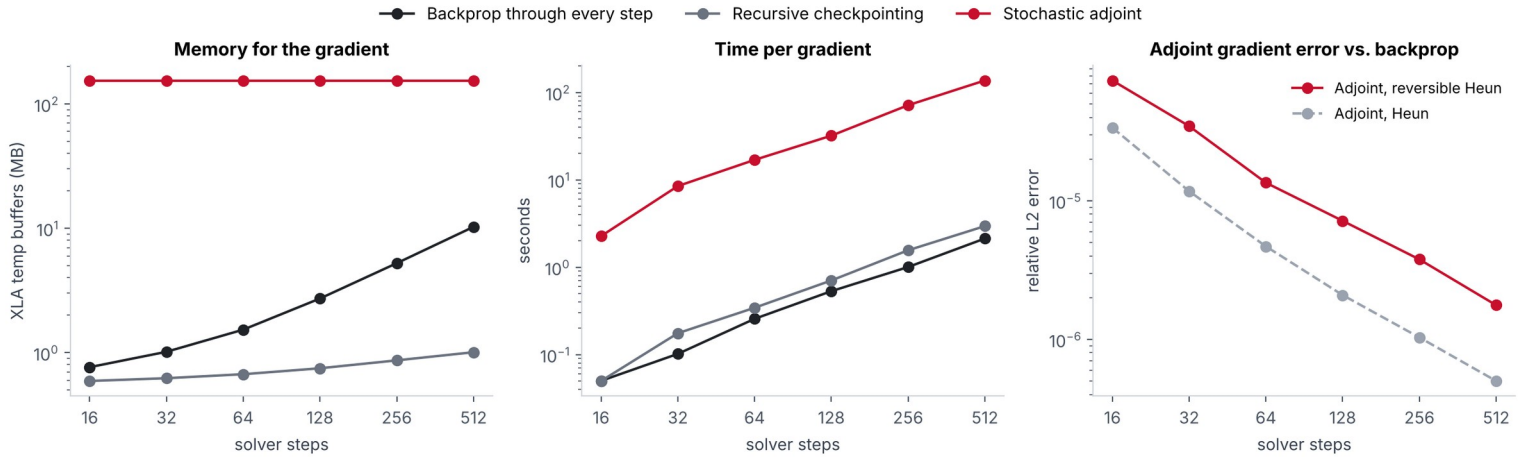
$$\frac{dL}{d\theta} = \int_0^T a_t^\top \partial_\theta \mu \, dt + \sum_j \int_0^T a_t^\top \partial_\theta \sigma_j \circ dW_t^j$$

Li, Wong, Chen & Duvenaud, AISTATS 2020. Why Stratonovich: A2.

Requirement	Solution	Cost
Same W forward and backward	virtual Brownian tree / Brownian Interval	O(1) memory, O(log 1/ε) per query
Chain rule must hold	Stratonovich form (convert Itô drift)	—
Exact gradients	reversible Heun + matching discrete adjoint (torchsde)	exact up to round-off

ODE adjoint + a noise term driven by the replayed Brownian path.

Three Ways to Get the Gradient, Measured



512 steps	Memory	Time / grad	Gradient
Backprop	10.3 MB	2.1 s	exact
Checkpointing	1.0 MB	3.0 s	exact
Adjoint	**154 MB**	**137 s**	rel. err 1.8e-6

16-dim neural SDE, 32 paths, reversible Heun, CPU. github.com/nishadw/neural-sde. Setup: A8.

In Diffrax, checkpointing beats the adjoint on memory, time and exactness at this scale.

Latent-SDE ELBO on the Avocado Data

prior (forecast)

$$\text{prior} \quad dZ_t = h_\theta(Z_t, t, c)\,dt + \sigma(Z_t, t, c)\,dW_t$$

posterior (filter, sees images)

$$\text{posterior} \quad dZ_t = h_\phi(Z_t, t, c, x_{0:K})\,dt + \sigma(Z_t, t, c)\,dW_t$$

objective

$$\log p(y_{1:K}) \geq \mathbb{E}_Q\left[\sum_k \log p(y_k \mid Z_{t_k})\right] - \mathrm{KL}_0 - \mathbb{E}_Q \int_0^T \tfrac{1}{2}\|u_t\|^2 dt$$

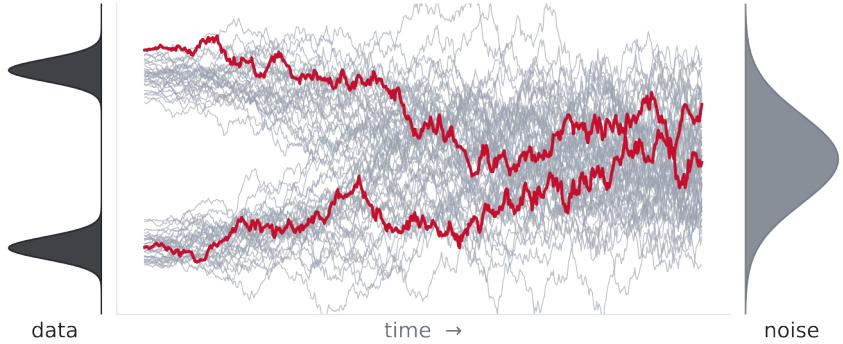
$$\sigma\, u_t = h_\phi - h_\theta$$

Li et al. 2020. KL derivation: A3.

ELBO term	Avocado data
y_k	stage label of image k
t_k	photo day (irregular)
c	T10 / T20 / Tamb
shared σ	else KL = ∞

Girsanov: path-space KL = running cost ∫½‖u‖²dt, integrated with the state.

Diffusion Models Are Neural SDEs



forward (fixed)

$$dx = f(x,t)\,dt + g(t)\,dW$$

reverse (learned score)

$$dx = \left[f(x,t) - g(t)^2\, s_\theta(x,t)\right] dt + g(t)\,d\bar{W}$$

probability-flow ODE

$$\frac{dx}{dt} = f(x,t) - \tfrac{1}{2}\, g(t)^2\, s_\theta(x,t)$$

denoising score matching

$$\min_\theta \; \mathbb{E}_{t,\,x_0,\,x_t} \| s_\theta(x_t,t) - \nabla_{x_t} \log p_t(x_t \mid x_0) \|^2$$

	Latent SDE (avocado)	Diffusion model
learns	μ, σ	score only
training	backprop through solver	simulation-free

Diffusion = neural SDE with fixed σ, trained simulation-free; ours learns σ and backprops through the solver.

Proposed Evaluation Against the Paper's Baseline

Shelf-life metric	Xavier et al. (CNN + linear)	Latent ODE	Latent SDE
MAE, T10 (days)	1.82	*TBD*	*TBD*
MAE, T20 (days)	0.75	*TBD*	*TBD*
MAE, Tamb (days)	0.54	*TBD*	*TBD*
MAE, all (days)	1.04	*TBD*	*TBD*
CRPS of τ, all (days)	1.04 (= MAE for a point)	*TBD*	*TBD*
80% interval coverage	n/a	*TBD*	*TBD*

Baseline: Table 6 (5-stage, by sample), Xavier, Rodrigues & Silva, Foods 13:1150 (2024). TBD = experiment in progress.

Same by-fruit split and MAE as the paper, plus calibration metrics that reward a distribution.

Same Structure Elsewhere

Domain	Latent state	Neural SDE role	Reference
Weather	atmosphere	diffusion sampling of ensembles	Price et al., Nature 2025 (GenCast)
Robotics	action trajectory	reverse diffusion / flow matching	Chi et al. 2023; Black et al. 2024
Quant finance	price, volatility	calibrated market models	Gierjatowicz 2022; Cohen 2021
Clinical	patient state	irregular vitals (ODE/CDE family)	Rubanova 2019; Kidger 2020
Generative models	image / audio	score-based sampling	Song et al. 2021

Wherever decisions depend on the distribution over trajectories.

Open Problems

Issue	Symptom	Evidence / mitigation
Adjoint error	backward re-solve ≠ forward path	ANODE (2019); 1e-4–1e-6 here
Stiffness	NFE grows during training	kinetic / Jacobian regularization (Finlay 2020)
Discontinuous targets	$\nabla\, 1\{\tau \le t^*\} = 0$ pathwise	bridge crossing, surrogate, score function
Identifiability	drift ↔ diffusion trade-off; latent reparam.	ordinal head anchors q; synthetic recovery test
Visual confounds	lighting, marks, wind rub look like ripening	paper: per-sample > per-picture accuracy
Uncontrolled covariates	Tamb 20% faster than Arrhenius	condition on humidity if logged

O(1) memory is real; exact, unbiased, identifiable gradients each cost extra.

DISCUSSION

Questions for the Room

1. Which estimator would you train $P(\tau \le t^*)$ with: surrogate, bridge crossing, or score function?

2. Slide 13 suggests extra spread at low temperature. How would you test whether σ depends on temperature?

3. Which differentiable simulator from this course would you make stochastic, and what happens to its gradients?

Code, benchmarks, slides: github.com/nishadw/neural-sde

Derivation: ODE Adjoint

1. Lagrangian with multiplier a(t) on ż = f

$$\mathcal{L} = L(z(T)) - \int_0^T a(t)^\top (\dot{z} - f_\theta(z,t))dt$$

2. Differentiate in θ; integrate $a^\top$ (d/dt)(∂z/∂θ) by parts

∂z/∂θ terms: [∂L/∂z(T) − a(T)]·∂z(T)/∂θ + ∫ (ȧ + $a^\top$ ∂f/∂z)·∂z/∂θ dt + a(0)$^\top$ ∂z(0)/∂θ

3. Set both brackets to zero

$$\frac{d\mathcal{L}}{d\theta} = \int_0^T a^\top \partial_\theta f \, dt \quad \text{when} \quad \dot{a} = -a^\top \partial_z f, \quad a(T) = \partial_z L$$

Add a(0)$^\top$ ∂z(0)/∂θ if the initial state depends on θ (e.g., z(0) = e_φ(x_0)).

Choose a(t) to cancel every ∂z/∂θ term; what remains is dL/dθ.

Why Stratonovich for the Stochastic Adjoint

$$dZ_t = \mu\, dt + \sigma \circ dW_t \quad \Longleftrightarrow \quad \dot{Z} = \mu + \sigma\, \dot{W} \quad (\text{ordinary chain rule})$$

	Itô	Stratonovich
integral evaluated at	left endpoint	midpoint
martingale property	yes	no
chain rule	Itô's lemma (+½ f″σ² dt)	ordinary
time reversal	extra correction terms	same form
drift conversion	μ	μ − ½ Σ_j (∂σ_j/∂z) σ_j

Stratonovich SDEs obey the ordinary chain rule, so the ODE adjoint carries over term by term.

Derivation: Path-Space KL via Girsanov

Girsanov (σu = h_φ − h_θ)

$$\frac{dQ}{dP} = \exp\left(\int_0^T u_t^\top \, dW_t^P - \tfrac{1}{2} \int_0^T \|u_t\|^2 dt \right)$$

Under Q: W^P = W^Q + ∫u dt; the Itô integral has zero mean

$$\mathrm{KL}(Q \,\|\, P) = \mathbb{E}_Q\left[\log \tfrac{dQ}{dP}\right] = \mathbb{E}_Q \int_0^T \tfrac{1}{2}\|u_t\|^2 dt$$

Conditions: same σσ$^\top$; Novikov: E_P exp(½∫‖u‖²dt) < ∞. **In code:** dℓ = ½‖u‖² dt appended to the state; ℓ_T is a one-sample MC estimate.

Same σσ$^\top$, different drift: log dQ/dP = stochastic integral + running cost.

Derivation: First-Passage Ratios (Slide 13)

density of τ for X = vt + σW, barrier at L > 0

$$f_\tau(t) = \frac{L}{\sigma\sqrt{2\pi t^3}} \exp\left(-\frac{(L - vt)^2}{2\sigma^2 t}\right)$$

⇒ E τ = L / v, Var τ = L σ² / v³ (inverse Gaussian, mean L/v, shape L²/σ²)

ratios between storage groups at equal distance L

$$\frac{\mathrm{sd}(\tau_{10})}{\mathrm{sd}(\tau_{20})} = \left(\frac{v_{20}}{v_{10}}\right)^{3/2} (\sigma \text{ fixed}), \qquad \frac{v_{20}}{v_{10}} \quad (\sigma^2 \propto v)$$

Rates from Table 4: v_{10} = 0.228, v_{20} = 0.473, v_amb = 0.518 stages/day. MAE used as a proxy for sd (assumes small CV); quantization predicts v_{20}/v too; labeling noise ignored.

Inverse-Gaussian first passage gives sd(τ) ∝ √(Lσ²/v³).

Arrhenius Fit (Slide 12)

Step	Expression	Value
rates	$v = 1 / \lvert\alpha\rvert$	0.228, 0.473, 0.518 d^{-1}
Q_{10}	v_{20} / v_{10}	2.07
E_a	$R \ln Q_{10} / (1/283.15 - 1/293.15)$	50.4 kJ/mol
v_amb predicted	$v_{20} \exp[-E_a/R\ (1/291.85 - 1/293.15)]$	0.431 d^{-1} (2.32 d/stage)
v_amb observed	1 / 1.929	**0.518 d^{-1} (1.93 d/stage)**
T implied	solve Arrhenius for v = 0.518	21.3 °C
Jensen (sd 1.2 °C)	$½ (E_a/RT^2)^2 \sigma_T^2$	< 1% effect

E_a from two rates; prediction at the ambient mean; implied temperature.

Brownian Bridges: Replay and Crossings

virtual Brownian tree: recursive bisection from a seed

$$W_s \mid W_u, W_v \sim \mathcal{N}\Big(W_u + \frac{s-u}{v-u}(W_v - W_u),\ \frac{(s-u)(v-s)}{v-u}\Big)$$

crossing probability within a step (σ_q = local vol of q, frozen)

$$\mathbb{P}(\text{cross } b \mid q_k, q_{k+1}) = \exp\Big(-\frac{2(b-q_k)(b-q_{k+1})}{\sigma^2 \Delta t}\Big)$$

P(τ ≤ t*) ≈ 1 − Π_k (1 − p_k), p_k = 1 if an endpoint ≥ b. $\sigma_q^2 = \|\nabla q(Z_k)^\top \sigma_\theta(Z_k)\|^2$. Drift cancels after conditioning on endpoints.

One conditional Gaussian regenerates W and gives a smooth crossing probability.

SDE Solvers

Solver	Converges to	Strong order	Weak order	Notes
Euler–Maruyama	Itô	½ (1 additive)	1	default
Milstein	Itô / Strat.	1	1	needs ∂σ; commutative noise if d > 1
Heun	Stratonovich	½ (general)	—	predictor–corrector
Reversible Heun	Stratonovich	½ (general)	—	algebraically reversible
SRA1 / ShARK	Stratonovich	1.5 (additive only)	—	state-independent σ

Kloeden & Platen (1992); Kidger et al. (2021); Diffrax docs.

Match the solver to the calculus you wrote and the order to the quantity you need.

Benchmark Setup (Slide 16)

Item	Setting
Model	16-dim state; drift + diagonal-diffusion MLPs, width 64, depth 2 (12.7k params)
Batch	32 paths as one 512-dim SDE, diagonal noise
Solver	reversible Heun; virtual Brownian tree tol 1/4096; float64; 2-core CPU
Memory	XLA compiled temp buffers
Adjoint memory scaling	256 → 1,024 channels: 62 → 438 MB; 4k → 42k params: 95 → 389 MB
Method's own state	2d + p ≈ 110 KB → rest is implementation constant

Trend robust; absolute numbers depend on hardware and size.

References and Figure Credits

Xavier, Rodrigues & Silva. Shelf-life management and ripening assessment of 'Hass' avocado using deep learning. Foods 13(8):1150, 2024. doi:10.3390/foods13081150. CC BY 3.0 (Figs. 1, 2, 6a, 11; Tables 3, 4, 6).

'Hass' Avocado Ripening Photographic Dataset. Mendeley Data, 2024. doi:10.17632/3xd9n945v8.

Chen et al. Neural ODEs. NeurIPS 2018.

Li, Wong, Chen, Duvenaud. Scalable Gradients for SDEs. AISTATS 2020.

Kidger, Foster, Li, Lyons. Efficient and Accurate Gradients for Neural SDEs. NeurIPS 2021.

Kidger. On Neural Differential Equations. PhD thesis, 2022.

Mohamed et al. Monte Carlo Gradient Estimation in ML. JMLR 2020.

Gholami, Keutzer, Biros. ANODE. IJCAI 2019.

Finlay et al. How to Train Your Neural ODE. ICML 2020.

Song et al. Score-Based Generative Modeling through SDEs. ICLR 2021.

Kloeden & Platen. Numerical Solution of SDEs. 1992.

Särkkä & Solin. Applied SDEs. CUP 2019.

Griewank & Walther. Revolve. ACM TOMS 2000.

Rubanova et al. 2019; Kidger et al. 2020 (CDEs); Price et al. 2025; Chi et al. 2023; Black et al. 2024; Gierjatowicz et al. 2022.